

6. [Maximum mark: 7]

In this question you may assume the result $\int x \operatorname{cosec}^2 x \, dx = \ln|\sin x| - x \cot x + c$.

Consider the differential equation $x \frac{dy}{dx} + 3y = \frac{\operatorname{cosec}^2 x}{x}$ for $0 < x < \pi$.

Given that $y = \frac{8}{\pi^2}$ when $x = \frac{\pi}{2}$, find the particular solution for this differential equation.

Express your answer in the form $y = f(x)$.

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